Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

Author Name
Affiliation
Address
email@example.com

Abstract

Conformal PID states long-run coverage for an arbitrary bounded scorecaster, and a published corollary retains its rate under any predictable perturbation of the deployed threshold inside an admissible radius. The opposite reading is in print too, under overlapping authorship, and acted on. We make two claims. The record holds both halves and we join them, conceding placement and derivation by name. And at the null scorecaster the corollary's condition is *tight*: a legal L_1 dead band on the *completed* threshold leaves that set past $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, which is the published radius itself there, and past it realised miscoverage is 1.000000 against $\alpha = 0.1$ while saturation fires every round. Leaving the set costs coverage and not only the rate: partial adjustment there overruns Proposition 2's bound $42\times$ at $T = 10^6$ and still covers, 0.100004. The edge belongs to the saturator–scorecaster pair, not to the readout, and elsewhere the condition is sufficient only: under the equally legal $\hat{q} \equiv +b/2$ the failing band covers, and under $\hat{q} \equiv -b/2$ both the failing band and partial adjustment at $w = 0.999$ return 1.000000.

1 Introduction

An online conformal method emits a threshold before each observation and moves it afterwards. Forecast stability names that movement, scores it and prices it [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying a forecast's temporal structure at fixed level was invented earlier, in forecast verification, where the predictive marginal is matched on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]; that coverage and mean length under-describe a deployed interval is established too [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

First: the record carries a claim and its refutation, and this paper joins them. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator and asks of it only that it be predictable and lie in $[-b/2, b/2]$ — "any function of the past" (Theorem 1) — so a readout placed in that slot is already quantified over. Also in print, of the same method: "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee" [Li and Rodríguez, 2025] — stated and acted on in that paper's baselines paragraph, refereed and verbatim on printed page 18443. The refutation is in print too, twice and with proof: Li et al. [2025, Corollary 2] add an arbitrary predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ to the deployed threshold, observe that the result "has exactly the CPID error-integration form", and conclude $|E_T| \leq c\, h(T) + 1 = o(T)$; the same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. The first and last authors of Li et al. [2025] are the authors of Li and Rodríguez [2025] and cite it: the halves already share a bibliography, this is ordinary self-correction, and the joining is what this paper adds. Four surfaces, queried on 20 August 2026 for the two papers' identifiers and for "scorecaster" — arXiv full text, arXiv abstracts, OpenReview, and the six works Semantic Scholar records as citing

Li and Rodríguez [2025] — return no document containing both the claim and the corollary. Hu et al. [2026] still calls scorecaster choice "arbitrary" and lacking "principled guidance", while asserting valid coverage: model selection, not validity. **We claim neither the placement nor the derivation.**

Second: that admissible set has an edge, and the condition there is tight. Corollary 2 is a *retention* result, so what it leaves open is outside that set. Section 3 exhibits a legal perturbation that leaves it — an L_1 dead band on the *completed* threshold, the family characterised here — and locates the edge at $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, the corollary's own radius at the null scorecaster. Past it the price is coverage and not only the rate, so the published sufficient condition is necessary there. Those two claims are the paper; Section 4 sets them beside four vocabularies for the same movement.

2 Setup

The producer. A forecaster fit on data before t induces a score s_t, a threshold q_t is emitted before y_t is revealed, the deployed set is $C_t = \{y : s_t(x_t, y) \le q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$; the target is long-run coverage, $T^{-1} \sum_{t \le T} \mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale,

$$q_{t+1} \;=\; \hat{q}_{t+1} \;+\; r_t\Big(\textstyle\sum_{i \le t}(\mathrm{err}_i - \alpha)\Big), \tag{1}$$

their iteration (5), in which r_t saturates: condition (4) requires $|r_t(x)| \ge b$ with the sign of x once $|x| \ge c\,h(t)$, for h nonnegative, nondecreasing and sublinear. Writing $E_t = \sum_{i \le t}(\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \le c\,h(T) + 1$; boundedness is read on realised scores.

One scalar, two readouts, two places to put it. Damping the movement of a sequentially updated scalar has two standard readouts, both prior work: a quadratic cost gives linear partial adjustment towards the target, $\tilde{q}_{t+1} = (1-\lambda)\hat{q}^{\mathrm{raw}}_{t+1} + \lambda \tilde{q}_t$ [Gârleanu and Pedersen, 2013], and a proportional cost a no-trade band, moving only when the target is more than τ away and then pulling back by exactly τ [Constantinides, 1986, Davis and Norman, 1990]. The first is published as post-processing [Godahewa et al., 2025] and applied to a deployed conformal threshold [Binny and Dixit, 2025, Eq. 13]. It has two positions in (1): **(B)** in the $\hat{q}_{t+1}$ slot, where both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, inside Corollary 2's admissible set; and **(A)** downstream of the completed q_{t+1}, outside it. Placement A is measured here. *The dead band (L1) is not a stylistic alternative to the partial-adjustment (L2) form*; it is the specific perturbation family whose failure §3 characterises. Neither is safe by construction: L2 forfeits Proposition 2's rate while covering, L1 past $\tau^\star$ loses coverage, and at the legal $\hat{q}_t \equiv -b/2$ both lose it. What is tight is the radius, not a form.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H} |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new: $\sum |\Delta q|$ over a predictive distribution is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it *jumpiness*, a *convergence index* and *(in)consistency* [Zsótér et al., 2009, Ehret, 2010, Pappenberger et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length. Table 2's caption names ten across four vocabularies, which is why a fresh one is used here.

Harness. $\alpha = 0.1$, $b = 2$, $c = 1$, $h(t) = \log(t+2)$, and the clipped saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, which meets condition (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 is stated about. Neither free choice is neutral: condition (4) is a lower bound this saturator meets exactly, and Theorem 1 permits scorecasters $b/2$ either side of the null one. The adversary is specified rather than tie-broken: it plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$ — the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold", firing the strict indicator. Legality is Theorem 1's $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.

3 The edge of the admissible set, measured

The retention result is published, and this section tests it from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \le \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and *retains* $|E_T| \le c\,h(T) + 1$; the admissible

Table 1: Placement A: a one-scalar readout on the *completed* threshold; one deterministic adversary, one pass to $T = 10^6$. *All eleven arms run are listed*, so no covering arm is withheld. Proposition 2's bound is 13.2061 at $T = 2\times10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, "Sat." the rounds on which (4) delivers its full $\pm b$.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t \|E_t\|$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	**0.100004**	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	**1.000000**	**1.000000**	900,000	60,747	1.0000	0.5

radius is theirs. A downstream partial adjustment of weight w leaves it. Its excursion at $T = 10^4$ is $6.30/63.00/623.70$ for $w = 0.9/0.99/0.999$, and for the last two it is *flat in* T while the bound grows like $\log T$, so the ratio falls with the horizon: at $w = 0.999$, 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$; the measured law across the arms is $\max_t |E_t| \approx \max\{0.5\, h(T) + 0.9,\ 0.63/(1 - w)\}$, and five of the eleven arms exceed the bound at $T = 10^6$. The forfeit is therefore a property of the smoothing *strength* and not of Placement A as such, and it buys something: the $w = 0.999$ arm cuts deployed travel from 28,311 to 202. The sign is the opposite of the intuitive one. A readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put condition (4) at risk; it delays the correction the integrator has already computed, so the accumulator excurses *further*. Smoothing the output does not damp E_t, it feeds it. The 623.70 is the primary regime's figure — the adversary landing $\varepsilon = 10^{-9}$ above the deployed threshold under the harness's strict indicator — and what it turns on is that ε rather than the counting convention: an adversary playing the threshold *exactly* returns 70.80 under the same indicator, because its signature move then covers.

The L_1 dead band leaves the radius, and the edge is measurable. A dead band pulls the deployed threshold back by exactly τ when it releases, so "inside Corollary 2" reads $\tau \le \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator's ceiling: its realised supremum is $b - \tau$ to the printed digit at every $\tau \ge 0.9$. The adversary's best legal play is $b/2$, so coverage is lost exactly when that supremum drops below it, at $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$. Every dead-band configuration in the variation sweep agrees, 19 of 19: three widths under each of the null scorecaster, $\hat{q} \equiv +b/2$, $\hat{q} \equiv -b/2$, a saturator at level $4b$ and Angelopoulos et al.'s tangent integrator, plus four wider bands at the null scorecaster. An eleven-point grid at the null scorecaster *locates* that edge between $\tau = 1.000$ and 1.001; the other four settings are bracketed at three widths and not located. **Here** $\sup r_t = b$ **and** $\hat{q} \equiv 0$**, so** $\tau^\star = b/2$ — one point of the admissible class, and both terms move it. At $\hat{q} \equiv +b/2$, legal, the same $\tau = 1.5$ band covers at $T = 10^6$ (0.100010, $\max_t |E_t| = 11.20$, inside the bound); at $\hat{q} \equiv -b/2$, equally legal, $\tau^\star = 0$ and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$. Raising the saturator's level to $4b$, admissible since (4) is a lower bound, raises $\sup_x r_t(x)$ and $\tau^\star$ with it and cuts 623.70 to 120.60; under the unbounded tangent integrator Angelopoulos et al. [2023] report using in all their experiments, $\sup_x r_t(x)$ is unbounded and so is $\tau^\star$, the three widths run there all cover, $\tau = 1.5$ at 0.100001, and 623.70 falls to 20.10, both at $T = 10^6$. **The method's own integrator therefore sits inside the set at every width run here.** Outside it the transition is a step: $\tau = 1.000$ covers at $T = 10^5$ while $\tau = 1.001$ returns 1.000000, and past $\tau^\star$ the threshold sits permanently below the reachable score range and $|E_t| = (1 - \alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$; $\tau = 2.0$, 2.5, 3.0 and 5.0 fail too at $T = 2\times10^5$, so the failing set is $(\tau^\star, \infty)$, open at the left. **The endpoint is the one row in the record where the adversary's convention decides the answer**: at $\tau = \tau^\star$ the deployed supremum equals the adversary's own cap $b/2$, so the primary regime covers while the exact-threshold play returns 1.000000 or 0.000000 according to the tie rule. Every other row is convention-free. **And the failing arms have** $\mathrm{Sat.} = 1.0000$: condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's other hypothesis, that the threshold closing the loop *is* the integrator's output.

What that measures is that the published condition is tight. At $\mu_t = 1$, $\hat{q}_t \equiv 0$ the measured $\tau^\star$ is exactly Corollary 2's radius, and crossing it costs coverage rather than the rate. **The claim is**

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	Forecast stability	Verification	Switching-cost OL
a distribution-free long-run-coverage guarantee for *any* bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2], and the admissible set measured here to be tight	the scored movement functional, the readout map as post-processing [Godahewa et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2], [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a control *stronger* than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T, which grows the ratio [Andrew et al., 2013]

that necessity: a tightness result about someone else's sufficient condition, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere the condition stays sufficient only — at $\hat{q}_t \equiv +b/2$ Corollary 2 permits radius 0 while every width run there covers — which is what makes "tight" a located claim.

The constructive reading, and it is the corollary's rather than ours. Placement B keeps $\{\hat{q}_t\}$ inside the admissible radius, so Theorem 1 carries it already; and at Placement A, condition (4) bounds $\sup_x r_t(x)$ only from below, so raising the saturator's level widens the band at no cost to the theorem. What the measurement adds is that the band has an edge, that the edge is computable before deployment from $\sup_x r_t(x)$ and $\sup_t \hat{q}_t$, and that crossing it is not a graceful degradation.

4 Where this sits

Four vocabularies name one quantity many ways, and the object that settles its validity — the admissible set around the deployed threshold — sits in the first of them.

What is conceded, by name. The placement is prior work: Angelopoulos et al. [2023] state coverage for any admissible integrator and any scorecaster and run a smoothing-family model in that slot themselves; Dupuy et al. [2026] give the generic argument in their Appendix A and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and that is the concession that matters most: Li et al.'s Corollary 2 proves the retention statement in Conformal PID's own notation [Li et al., 2025], with Wang and Hyndman [2024] an independent second instance. The smoother object, the metric arithmetic and both readout maps are conceded in Section 2, and Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path, where free-until-the-bound-binds is the defining specification rather than a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score; their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha\, h(T))$ for a sublinear h, supported with a plot rather than derived. That accumulation is what Section 3 measures diverging.

5 Limitations

One construction, one adversary, one saturator, and $0.63/(1-w)$ is specific to the saturator's *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, so the certified gap $(c\, h(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$: what is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\, b$; under i.i.d. scores it returns 0.249376 at $T = 10^6$, which is not a retreat but an adversary-free confirmation of the mechanism — the threshold pins at $b - \tau = 0.5$ and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. One seed, and only the null scorecaster's edge is located**; the other four settings are bracketed at three widths. **What would overturn the claim** is a legal $(r_t, \hat{q}_t)$ pair with $\tau \leq \tau^\star$ that loses coverage, or $\tau > \tau^\star$ that keeps it; the sweep has neither.

References

Lachlan L. H. Andrew, Siddharth Barman, Katrina Ligett, Minghong Lin, Adam Meyerson, Alan Roytman, and Adam Wierman. A tale of two metrics: Simultaneous bounds on competitiveness and regret. In *Proceedings of the 26th Conference on Learning Theory (COLT 2013)*, pages 741–763, 2013.

Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal pid control for time series prediction. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. doi: 10.52202/075280-1000.

Allen Emmanuel Binny and Anushri Dixit. Who moved my distribution? conformal prediction for interactive multi-agent systems. arXiv:2511.11567v1, 2025.

Jiadong Chen, Yang Luo, Xiuqi Huang, Fuxin Jiang, Yangguang Shi, Tieying Zhang, and Xiaofeng Gao. IPOC: An adaptive interval prediction model based on online chasing and conformal inference for large-scale systems. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 202–212, 2023. doi: 10.1145/3580305.3599396.

George M. Constantinides. Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862, 1986. doi: 10.1086/261410.

M. H. A. Davis and A. R. Norman. Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713, 1990. doi: 10.1287/moor.15.4.676.

Ricarda Duerst, Jonas Schöley, Julia Hellstrand, and Mikko Myrskylä. Calibrating probabilistic forecast paths on past forecast errors: an application to the Finnish total fertility rate. Technical Report WP-2024-016, Max Planck Institute for Demographic Research, 7 2024.

Théo Dupuy, Binbin Xu, Stéphane Perrey, Jacky Montmain, and Abdelhak Imoussaten. Relevance-aware thresholding in online conformal prediction for time series. In *Communications in Computer and Information Science*, pages 196–217, 2026. doi: 10.1007/978-3-032-16708-8_17.

Uwe Ehret. Convergence index: a new performance measure for the temporal stability of operational rainfall forecasts. *Meteorologische Zeitschrift*, 19(5):441–451, 2010. doi: 10.1127/0941-2948/2010/0480.

Sergio Galeani, Sophie Tarbouriech, Matthew Turner, and Luca Zaccarian. A tutorial on modern anti-windup design. *European Journal of Control*, 15(3-4):418–440, 2009. doi: 10.3166/ejc.15.418-440.

Ben Gao, Jordan Patracone, Stéphane Chrétien, and Olivier Alata. Conformal online learning of deep Koopman linear embeddings. arXiv:2511.12760v2, 2025.

Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013. doi: 10.1111/jofi.12080.

Evgenii Genov, Julian Ruddick, Christoph Bergmeir, Majid Vafaeipour, Thierry Coosemans, Salvador García, and Maarten Messagie. Balancing forecast accuracy and switching costs in online optimization of energy management systems. *Expert Systems with Applications*, 298:129305, 2026. doi: 10.1016/j.eswa.2025.129305.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2): 243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

Rakshitha Godahewa, Christoph Bergmeir, Zeynep Erkin Baz, Chengjun Zhu, Zhangdi Song, Salvador García, and Dario Benavides. On forecast stability. *International Journal of Forecasting*, 41(4): 1539–1558, 2025. doi: 10.1016/j.ijforecast.2025.01.006.

Dongjian Hu, Junxi Wu, Shu-Tao Xia, and Changliang Zou. Distribution-informed online conformal prediction. arXiv:2512.07770v2, 2026.

Ankit Lade, Sai Krishna J., and Indar Kumar. Bias-corrected adaptive conformal inference for multi-horizon time series forecasting. arXiv:2604.13253v1, 2026.

Ruipu Li and Alexander Rodríguez. Neural conformal control for time series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 18439–18447, 2025. doi: 10.1609/aaai.v39i17.34029.

Ruipu Li, Daniel Menacho, and Alexander Rodríguez. Optimization-based online conformal prediction for multi-step forecasting. arXiv:2508.13362v3, 2025.

Yizhou Min, Yizhou Lu, Lanqi Li, Zhen Zhang, and Jiaye Teng. Questioning the coverage-length metric in conformal prediction: When shorter intervals are not better. arXiv:2601.21455v2, 2026.

F. Pappenberger, H. L. Cloke, A. Persson, and D. Demeritt. HESS Opinions "on forecast (in)consistency in a hydro-meteorological chain: curse or blessing?". *Hydrology and Earth System Sciences*, 15(7):2391–2400, 2011. doi: 10.5194/hess-15-2391-2011.

P. Pinson and R. Girard. Evaluating the quality of scenarios of short-term wind power generation. *Applied Energy*, 96:12–20, 2012. doi: 10.1016/j.apenergy.2011.11.004.

Kandrika Pritularga and Nikolaos Kourentzes. Forecast congruence: A quantity to align forecasts and inventory decisions. SSRN working paper 4711817, 2024.

Pouria Sarhadi. On the standard performance criteria for applied control design: PID, MPC or machine learning controller? arXiv:2503.14379v3, 2025.

Huseyin Tunc, Onur A. Kilic, S. Armagan Tarim, and Burak Eksioglu. A simple approach for assessing the cost of system nervousness. *International Journal of Production Economics*, 141(2): 619–625, 2013. doi: 10.1016/j.ijpe.2012.09.022.

Jente Van Belle, Ruben Crevits, and Wouter Verbeke. Improving forecast stability using deep learning. *International Journal of Forecasting*, 39(3):1333–1350, 2023. doi: 10.1016/j.ijforecast.2022.06.007.

Jente Van Belle, Honglin Wen, Wouter Verbeke, and Pierre Pinson. Stabilizing distribution-free probabilistic forecasts. arXiv:2605.28531v1, 2026.

Rahul Vaze. Simultaneous coverage and efficiency guarantee in online conformal prediction. arXiv:2607.26577v1, 2026.

Xiaoqian Wang and Rob J. Hyndman. Online conformal inference for multi-step time series forecasting. arXiv:2410.13115v2, 2024.

Rochelle P. Worsnop, Michael Scheuerer, Thomas M. Hamill, and Julie K. Lundquist. Generating wind power scenarios for probabilistic ramp event prediction using multivariate statistical post-processing. *Wind Energy Science*, 3(1):371–393, 2018. doi: 10.5194/wes-3-371-2018.

Junxi Wu, Dongjian Hu, Yajie Bao, Shu-Tao Xia, and Changliang Zou. Error-quantified conformal inference for time series. arXiv:2502.00818v2, 2025.

Marco Zanotti. Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems. arXiv:2506.05776v3, 2026.

Mingyi Zheng, Hongyu Jiang, Yizhou Lu, and Jiaye Teng. Smoothing-based conformal prediction for balancing efficiency and interpretability. arXiv:2509.22529v1, 2025.

Ervin Zsótér, Roberto Buizza, and David Richardson. "Jumpiness" of the ECMWF and Met Office EPS control and ensemble-mean forecasts. *Monthly Weather Review*, 137(11):3823–3836, 2009. doi: 10.1175/2009MWR2960.1.